Dijkstra’s Shortest Path Algorithm

Topic: Greedy File: g5.pdf

PROBLEM

Find shortest-path distances from a source vertex in a graph whose edge weights are non-negative.

GREEDY CHOICE

Repeatedly finalize the unvisited vertex with the smallest tentative distance, then relax its outgoing edges.

ALGORITHM

Set dist[source] = 0 and all other distances to infinity. Use a min-priority queue; for each edge u->v, update dist[v] if dist[u] + weight(u,v) is smaller.

EXAMPLE

If S-A costs 2, S-B costs 7, and A-B costs 1, the shortest distance from S to B is 3 via A.

COMPLEXITY

With adjacency lists and a binary heap: O((V + E) log V) time and O(V + E) graph storage.

IMPORTANT

Dijkstra is not correct for graphs with negative edge weights. Consider Bellman-Ford for those cases.